Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A sample warms by **3°C** after a certain energy input. If absorbed energy becomes **5** times as large with the same mass and material and no phase change, find the new rise.

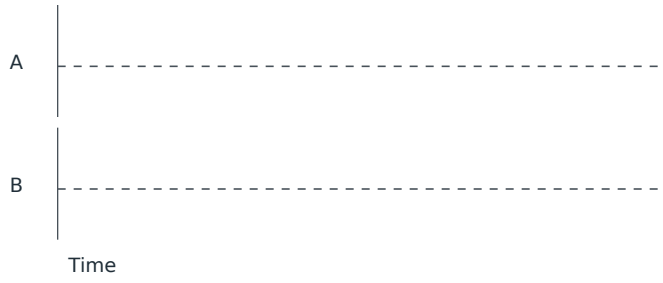
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: f = cycles/Δt; T = **1**/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes **6 cycles** in **3 s**: f=**2 Hz** and period T=**0.5 s**. Amplitude measures maximum displacement, not the spacing of crests.

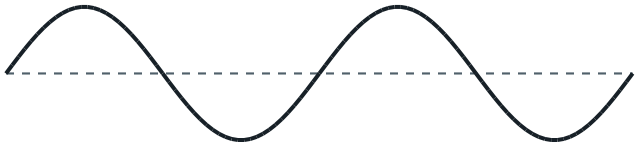
Your turn: Draw one spatial wave and **label amplitude** and wavelength **0**. Separately mark one period on a displacement-time graph.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Label amplitude** and one wavelength on the wave diagram.
2. **Explain why crest-to-trough height** is not the amplitude.

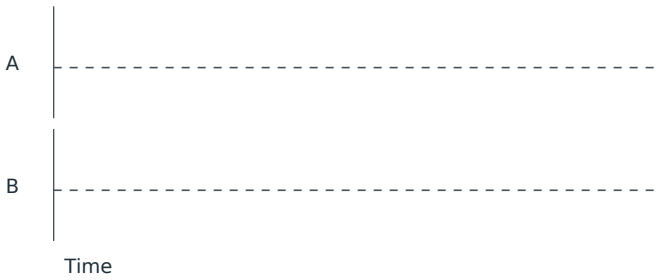


Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A point on a rope completes **6 cycles** in **5 s**. **Find its frequency** and period. In a separate trial, each cycle takes **0.5 s**. Find that trial's frequency. **Use two time** traces to **show which trial oscillates more** often.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency? Use the definitions to explain **0**.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** sample has c=**600 J**/(kg·°C) and warms by **8°C** without changing phase. **Find Q**.
Label temperatures, samples and relevant energy transfers.

Sample A

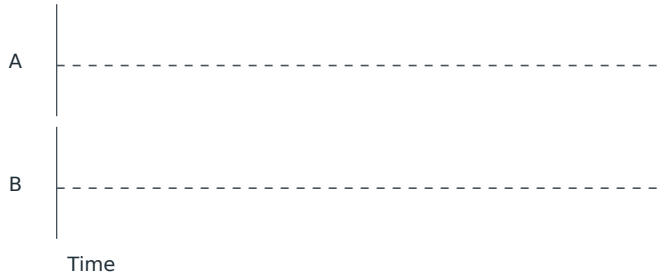
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

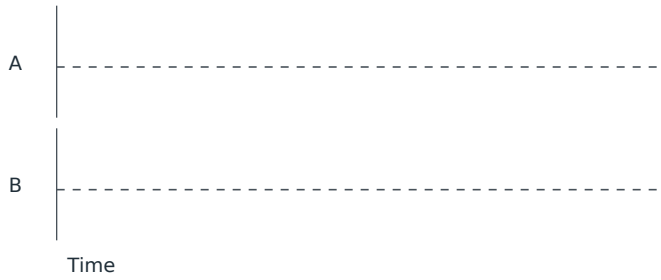
A wave has period **2 s**. **How many** cycles occur in **6 s**? **Find frequency**.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Draw two waves** with equal wavelength but different amplitudes.
2. **Explain which feature changes** and which does not.



Marzano check - circle: 1 2 3